PRACTICE SHEET

1. If $f(x) = A \sin$ and $\int_0^1 f(x)\,dx = \frac{2A}{\pi}$, then what is the value of B?

(a) $2/\pi$ (b) $4/\pi$
(c) 0 (d) 1

2. If m and n are integers, then what is the value of $\int_0^{\pi} \sin mx \sin nx\,dx$, if $m \neq n$?

(a) 0 (b) $\frac{1}{m+n}$
(c) $\frac{1}{m-n}$ (d) mn

3. What is the value of $\int_0^1 (x-1)e^{-x}\,dx$?

(a) 0 (b) e
(c) $\frac{1}{e}$ (d) $\frac{-1}{e}$

4. $\int_{\ln 2}^{x} (e^x - 1)\,dx = \ln\frac{3}{2}$, then what is the value of x?

(c) $\frac{1}{112}$ (d) $\frac{1}{119}$

8. What is the value of $\int_{-1}^{1} x\,|x|\,dx$?

(a) 2 (b) 1
(c) $\frac{1}{4}$ (d) 0

9. What is the value of $\int_0^1 \cos^8 x dx$?

(a) $\frac{35\pi}{256}$ (b) $\frac{70}{256}$
(c) $\frac{16}{35}$ (d) $\frac{8\pi}{35}$

10. What is $\int_a^b \frac{\log x}{x}\,dx$ equal to?

(a) (1, 2) log (ab). Log (b/a)
(b) log b/log a
(c) log (b/a)
(d) (1/2) log [(a + b)/ab]

11. What is the value of $\int_0^1 xe^{x^2}\,dx$?

(a) $\frac{(e-1)}{2}$ (b) $e^2 - 1$